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Key factors of mercury methylation and methylmercury accumulation in paddy soils

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ABSTRACT

Rice consumption serves as a primary pathway for human dietary exposure to methylmercury (MeHg), a potent neurotoxin predominantly formed in paddy soils. Although soil physicochemical properties exert a strong influence on MeHg production and accumulation, soil heterogeneity has hindered the identification of consistent controlling factors. In this study, a systematic meta-analysis of 22 peer-reviewed datasets was conducted to evaluate environmental regulators of MeHg contamination in paddy soils, with a specific distinction made between sites located within and outside Hg mining areas. The results revealed significantly elevated Hg and MeHg levels in paddy soils associated with Hg mining activity. Across both site types, soil pH, organic matter (OM), and total Hg (THg) were identified as key determinants of MeHg dynamics. MeHg accumulation was highest under weakly alkaline conditions (pH 7.0–8.0), while both low (<10 g/kg) and high OM (>60 g/kg) OM levels were unfavorable for MeHg accumulation. In Hg mining area paddy fields, maximum %MeHg occurred at OM levels of 30–40 g/kg, whereas in ordinary paddy fields, the optimal range was 20–30 g/kg. In addition, rice cultivation significantly absorbs soil MeHg. In acidic paddy soils, modest pH reduction coupled with appropriate OM amendments may suppress MeHg production. These findings offer important insights for the targeted management and remediation of Hg and MeHg in paddy soil.

1. Introduction

Mercury (Hg), a globally distributed heavy metal, actively participates in biogeochemical cycling via atmospheric transport and oceanic circulation. Through biotic and abiotic transformations, Hg is converted to MeHg, a potent neurotoxin via the methylation process (Schaefer et al., 2011). Hg methylation in paddy soils is a significant environmental concern due to the potential formation of MeHg and its accumulation in rice, posing substantial health risks (Chang et al., 2020; Xie et al., 2021).

Increasing studies have demonstrated notable MeHg accumulation in both paddy soils and grain tissues (Brombach et al., 2017; Liu et al., 2019a; Meng et al., 2011; Rothenberg, 2016a). Hg methylation in soil is regarded as the main source of MeHg in rice (Liu et al., 2023; Qin et al., 2018). A comprehensive analysis of nearly 60 years of global data highlights rice as an important dietary source of MeHg exposure, especially in Southeast and South Asia (Liu et al., 2019b). The paddy fields in Hg mining areas serve as critical sentinel sites for MeHg research due to certain Hg input. Spatial gradient analysis reveals significantly declining soil Hg and MeHg concentrations with increasing distance from Hg mining areas (Abeysinghe et al., 2017). Non-mining areas paddy fields (hereafter referred to as ordinary paddy) exhibit detectable Hg and MeHg contamination, although the degree of pollution is far less than Hg mining area paddy fields, demonstrating ubiquitous methylation potential across agricultural landscapes (Liu et al., 2019c). Therefore, understanding the similarities and differences in MeHg pollution status between in Hg mining areas paddy fields and ordinary paddy fields is of crucial importance for the classification management of paddy fields and public food safety.

Multiple environmental factors, including moisture, temperature, total mercury (THg), pH, organic matter (OM), total nitrogen (TN), and total sulfur (TS) (Du et al., 2021; Li et al., 2023a; Liu et al., 2014, Liu et al., 2019c; Zhou et al., 2022), have been reported to influence MeHg production and accumulation in paddy soil. Zhou et al. (2022) found

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that flooding was beneficial to Hg methylation in paddy fields, and the increase of temperature could promote this process by enhancing the metabolism of microbial communities (Zhou et al., 2022). Carbon, nitrogen and sulfur, as the fundamental elements in cropland ecosystems, can significantly affect soil chemical properties, thus affecting soil Hg bioavailability and methylation (Li et al., 2023b). An important study showed that the addition of C alone increased MeHg production by about 2–13 fold in paddy soils, whereas the co-addition of N and C markedly attenuated this stimulatory effect (Yan et al., 2019). Moreover, both elemental sulfur and sulfate were shown to elevate MeHg production by 2.4–5.7 fold (Li et al., 2023a). Emerging evidence also indicates that microplastics may facilitate MeHg formation in soils (Hao et al., 2023).

Recently, acidic/alkaline organic matter such as biochar amendments are often applied for the mitigation of Hg contamination in paddy fields using the function of pH and OM (O'Connor et al., 2018; Xing et al., 2020). Increasing studies have shown that pH and OM are associated with MeHg production and accumulation in paddy soil (Eckley et al., 2021; Jeong et al., 2024; Schartup et al., 2013). However, due to the heterogeneity of the soil, the effects of pH and OM on MeHg production and accumulation remains inconsistent. For instance, a negative correlation between soil pH/OM and MeHg content was reported (Abeysinghe et al., 2017; Yin et al., 2018). Liu et al. (2019c) research showed soil MeHg content was positively correlated with soil pH but was negatively correlated with OM across different paddy soil type in China (Liu et al., 2019c). Zhang's study showed that there was no significant correlation between MeHg content and OM or pH, instead identifying a positive correlation with THg content, which varied depending on the distances from Hg mining area to paddy soils (Zhang et al., 2010). Furthermore, OM incorporation can promote both the production of MeHg in soil (Li et al., 2023b). Dissolved OM has been demonstrated to enhance MeHg production effectively (Hu et al., 2023; Yang et al., 2023). However, a study reported that OM and its dissolved components can limit the bioavailability of Hg, thereby reducing the production of MeHg in paddy soil (Abdelhafiz et al., 2023). The heterogeneous nature of soils and environments conditions complicates the establishment of consistent, quantitative conclusions regarding the impact of geochemical factors on the Hg methylation and MeHg accumulation in paddy soils. This uncertainty hinders advancements in developing technologies and strategies to mitigate the risks associated with Hg contamination in paddy soils.

To elucidate the environmental drivers of Hg methylation and MeHg accumulation in paddy soils, we conducted a systematic review of the current literature and performed a meta-analysis to quantify the effects of two key factors pH and OM. Specifically, this analysis not only distinguished MeHg and its associated drivers between paddy fields in mercury mining areas and those in non-contaminated regions, but also considered the potential influence of rice cultivation on soil MeHg dynamics. Among the various variables investigated, pH and OM emerged as the most extensively studied and consistently factors influencing MeHg dynamics. These factors not only dominate experimental studies but also represent critical *in situ* parameters governing Hg and MeHg behavior in paddy soil. As such, pH and OM constitute primary targets for filed-scale mitigation and management strategies aimed at reducing MeHg production and accumulation in paddy soil.

2. Materials and methods

2.1. Literature collection and screening

The data for this study were obtained from major English-language databases, including Web of Science, Elsevier, and Google Scholar. The primary search key words included: "mercury", "methylmercury", "methylation", "paddy soil", and "mercury mining area". Although early investigations into this topic emerged during the 1990s, research activity remained limited until a marked increase in scholarly attention occurred after 2010 (Meng et al., 2018; Zhang et al., 2023). Using the key words "MeHg" and "paddy field", we conducted a comprehensive search of publications from 2001 to 2024 across multiple databases. The results revealed a sharp increase in the number of relevant studies beginning in 2010 (Fig. S1). This trend is probably attributable to the extensive development and broader application of MeHg analytical technique around 2010.

To ensure data quality, we systematically screened the retrieved literature from the period between January 2010 to June 2024. Duplicate entries, including those resulting from variation in spelling or metadata formatting, were removed. A total of 3038 articles were identified, comprising 689 from Web of Science, 332 from Elsevier, and 2017 from Google Scholar. The following inclusion criteria were subsequently applied to select studies for meta-analysis.

(1) The research objects are Hg mining area paddy fields or ordinary paddy fields;
(2) At a minimum, included studies were required to report key physicochemical parameters of the sampled paddy soil, including soil pH, OM or soil organic carbon (SOC) content, THg content and MeHg concentration. Moreover, the methodologies used to determine these parameters had to be clearly described within the respective publications;
(3) All data were derived from *in situ* paddy field soil, with samples reported to be properly collected and preserved to ensure data integrity;
(4) The primary response variable was MeHg concentration in paddy soils. For each dataset, a minimum of three replicates was required, with results reported as the mean ± standard deviation (SD) or mean ± standard error (SE).

Based on the established literature screening criteria, a total of 18 studies were ultimately included (Table S1). The detailed screening and exclusion process are illustrated in Fig. S2. Furthermore, to investigate the effects of key environmental factors on the rate of Hg methylation in paddy soils, the keywords "Hg methylation", "paddy soils", and "rice" were used in the literature search. The inundation incubation experiments for Hg-contaminated paddy soils were categorized based on the presence or absence of rice cultivation, ensuring that the included studies only varied in incubation conditions. The final screening resulted in 7 studies (Table S2), which were categorized based on the presence or absence of rice cultivation. Notably, two key studies involving exogenous substance amendments-one investigating the effect of C/N/S inputs on soil MeHg (Li et al., 2023b) and the other focusing on the connection between microplastics and MeHg production in paddy fields (Hao et al., 2023) -were included in both datasets. To ensure consistency in environmental context and incubation conditions across studies, data from treatment involving exogenous substances were excluded. Consequently, only the control group data from these two studies were included.

2.2. Statistical analysis and data processing

A portion of the study data was obtained by estimating graphical data using the Engauge Digitizer 4.1 software (Wojtyniak et al., 2020). To ensure the rigor and accuracy of the study, an examination of data distribution was conducted prior to performing the combined calculations. To mitigate the influence of extreme values and sample sizes on the combined effect, potentially distorting the final results, the study employed the BootES package in the R statistical software to normalize the samples (Kirby et al., 2013). Furthermore, means and 95 % confidence intervals for meta-analyses were calculated using the nonparametric bootstrap method (version 3.3.2) within the BootES package of the R software. The mean effects and 95 % confidence intervals (95 % CI) for various groups or subgroups were determined by iterating the bootstrapping procedure 999 times (Kirby et al., 2013). Single-factor

correlations were assessed via Spearman analysis, while significant differences between samples were evaluated using t-tests in IBM SPSS Statistics 26. Following standard numerical conversion approaches commonly employed in meta-analysis, soil pH, OM, and SOC content were normalized using the following procedures (Liu et al., 2016; Nguyen et al., 2017; Tian et al., 2023):

$$pH(H_2O) = 1.65 + 0.86 \times pH(CaCl_2) \quad (1)$$

$$OM = \frac{SOC}{0.58} \quad (2)$$

3. Result

3.1. Hg pollution and influencing factors of Hg accumulation in paddy soil

The physicochemical properties of the soil as well as the extent of Hg contamination exhibited statistically significant differences between Hg mining area paddy soils and ordinary paddy soils ($p < 0.001$, Table S3). The concentrations of THg and MeHg in the paddy soils derived from the Hg mining area were markedly elevated compared to those found in ordinary paddy fields, with THg ranging from 0.24 to 990 μg/g and MeHg concentrations spanning from 0.03 to 315.16 ng/g, as opposed to the respective values of 0.02–1.23 μg/g and 0.24–4.70 ng/g documented in the paddy fields (Fig. 1a and b). Conversely, paddy soils located in the Hg mining area, characterized by elevated content of THg and MeHg, demonstrated a significantly reduced %MeHg (Fig. 1a and b).

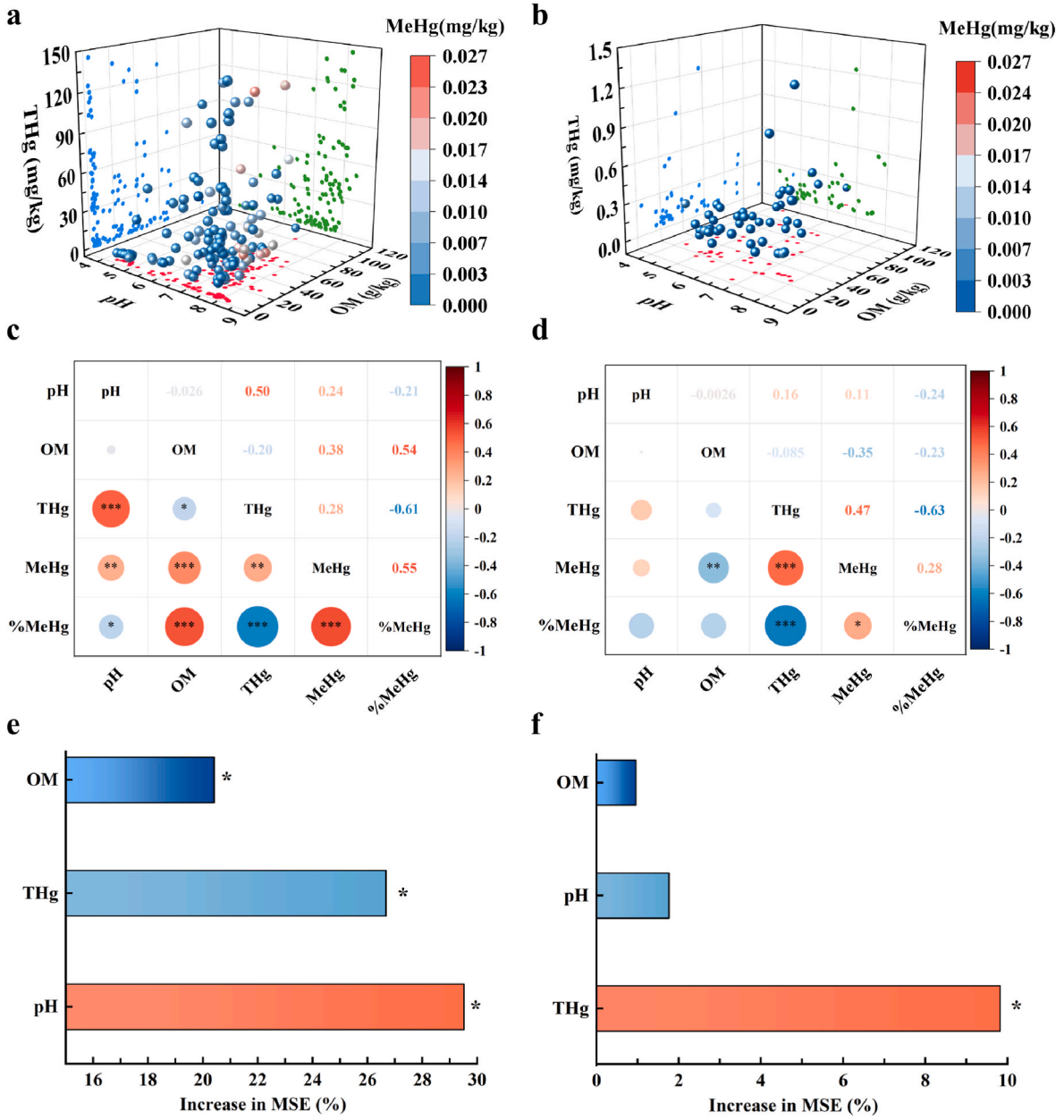


Fig. 1. Distribution of key physicochemical indicators in paddy soils from the Hg mining area (a) and ordinary paddy soil (b). Correlation analysis of these key physicochemical indicators in Hg mining area paddy soils (c) and in ordinary paddy soils (d). Importance of these indicators for MeHg concentration by a random forest model in the Hg mining area paddy soils (e) and ordinary paddy soils (f).

Furthermore, our analysis revealed that the concentration of MeHg exhibited a positive correlation with soil pH, OM and THg, while %MeHg was found to be negatively correlated with pH and positively correlated with OM in paddy soils derived from Hg mining areas ($p < 0.01$, Fig. 1c). In ordinary paddy fields, both soil OM and THg exhibited a positive correlation with MeHg concentration ($p < 0.05$, Fig. 1d). We employed a random forest model to identify the significant environmental variables influencing MeHg accumulation. Our findings indicated that THg were critical determinants of MeHg content in both types of paddy fields. However, soil pH and OM values only have significant effects on the accumulation of MeHg in Hg mining area paddy fields ($p < 0.05$, Fig. 1e and f).

3.2. Accumulation of MeHg in paddy soils with different pH and OM

Both paddy soil systems demonstrated congruent MeHg accumulation patterns under pH and OM gradients, exhibiting characteristic biphasic responses in %MeHg dynamics with well-defined optima (Fig. 2). Methylation efficiency peaked in weakly alkaline environment (pH 7.0–8.0) across both soil types, while lower OM environments (<10 g/kg) substantially constrained %MeHg. The %MeHg in both the Hg mining and ordinary paddy soils displayed a progressive increase with elevating pH values, attaining its maximum at pH 7.0 to 8.0 (Fig. 2a–Table S4). Simultaneously, both paddy soils demonstrated markedly low %MeHg when OM content was below 10 g/kg, while the %MeHg in the paddy soils reached a peak with increasing OM content (Fig. 2b–Table S5). Notably, as OM content increased, %MeHg in paddy fields decreased significantly after reaching the peak (Fig. 2b–Table S5). However, due to the difference between the two types of paddy fields, %MeHg peaked at 30–40 g/kg of OM content in Hg mining area paddy fields and at 20–30 g/kg in ordinary paddy fields (Fig. 2b). This might be attributed to the relatively limited availability of OM in paddy fields within Hg mining areas, where sustained Hg methylation depends on increased OM input (Haitzer et al., 2002; He et al., 2019).

3.3. Hg methylation in paddy soils at different pH, OM, THg

Given that the soil was subjected to both rice cultivation and fallow periods throughout the paddy cultivation cycle, we investigated the production and accumulation of MeHg in paddy soils under with/without cultivation conditions to assess the effects of pH, OM, and THg. The literature was carefully selected and 7 literatures were obtained (Fig. S2, Table S2). Our analysis revealed that rice cultivation exerted a significant influence on the accumulation of MeHg in paddy soils. The increase in %MeHg was more pronounced in soils lacking rice cultivation compared to those with rice cultivation ($p < 0.05$, Fig. S5), indicating that rice cultivation significantly mitigates the accumulation of newly produced MeHg in soils and suggesting that soil Hg methylation serves as the primary source of MeHg in rice. In the absence of rice cultivation, a lower OM environment was more conducive to %MeHg production and exhibited significant Hg methylation in weakly acidic conditions alongside minimal Hg input (Fig. 3a and b). In the soil for rice cultivation, when the THg content in the soil is greater than 10 mg/kg, the change in %MeHg in the paddy soil increases by nearly 15 times on average (Fig. 3c). Furthermore, linear regression analysis indicated that %MeHg change was positively correlated with pH and negatively correlated with OM content in the absence of rice cultivation ($p < 0.05$, Fig. 3g and h). Conversely, a positive correlation was observed between THg content and %MeHg change in rice cultivation areas ($p < 0.01$, Fig. 3f).

4. Discussion

4.1. Response of MeHg accumulation in paddy fields to environment

Our findings reveal significantly elevated Hg and MeHg contamination loads in the Hg mining area paddy fields compared to agricultural control systems. A large number of Hg-containing substances are released into the environment during the mining process and enter the paddy fields through means such as atmospheric deposition and underground migration (Qiu et al., 2008; Xu et al., 2018). However, surprisingly, compared with in the Hg mining area paddy fields with a higher total Hg content, the common paddy fields have lower MeHg and higher %MeHg, which may be related to the availability of Hg in the soil. A large number of studies have shown that inactive mineral Hg, namely "old Hg", is the main form of Hg in Hg mining areas paddy field soils (such as cinnabar), while anthropogenic emissions are the main source of Hg in ordinary paddy field soils, namely "new Hg" (Meng et al., 2014; Yin et al., 2016; Zhu et al., 2015). Compared with "old Hg", "new Hg" may have higher bioavailability (Jonsson et al., 2014; Liu et al., 2019c). Therefore, it is very necessary to monitor the Hg content in the surrounding environment of Hg mining areas and limit the Hg input from ordinary paddy fields.

Our meta-analysis elucidates linkages between pH/OM gradients and MeHg accumulation patterns in rice paddies, providing actionable strategies for *in situ* physicochemical remediation of contaminated agroecosystems. Consistent with existing studies, the accumulation of MeHg in paddy fields is strongly associated with pH and OM, indicating that in-situ remediation of severely MeHg-contaminated paddy fields can be carried out subsequently by adjusting soil acidity and alkalinity and adding biomass. The high levels of %MeHg in Hg mining area paddy fields and ordinary paddy fields in a weakly alkaline environment may

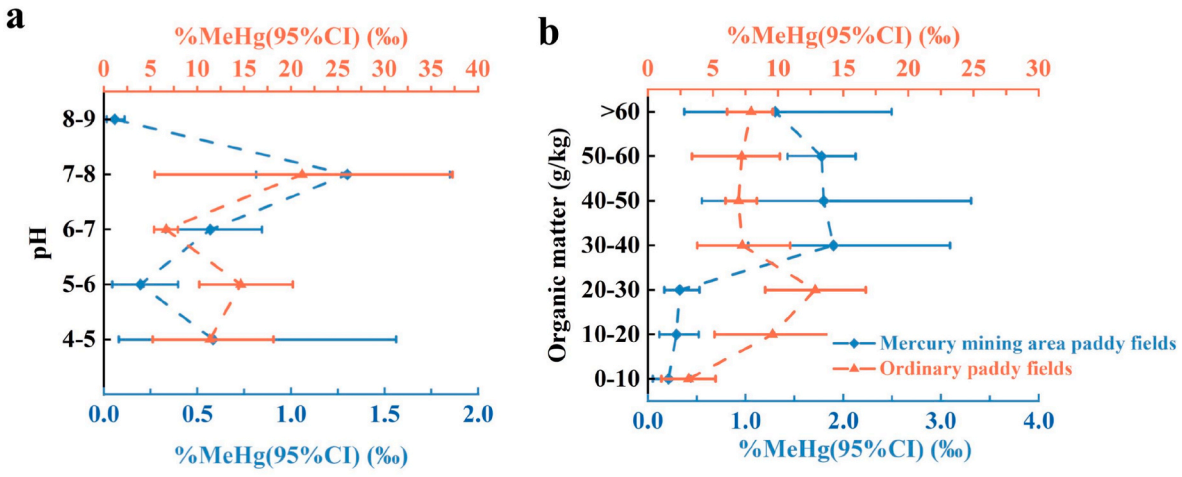


Fig. 2. Distribution and variation of %MeHg with varying pH (a) and organic matter content (b) in both Hg mining area paddy soils and ordinary paddy soils.

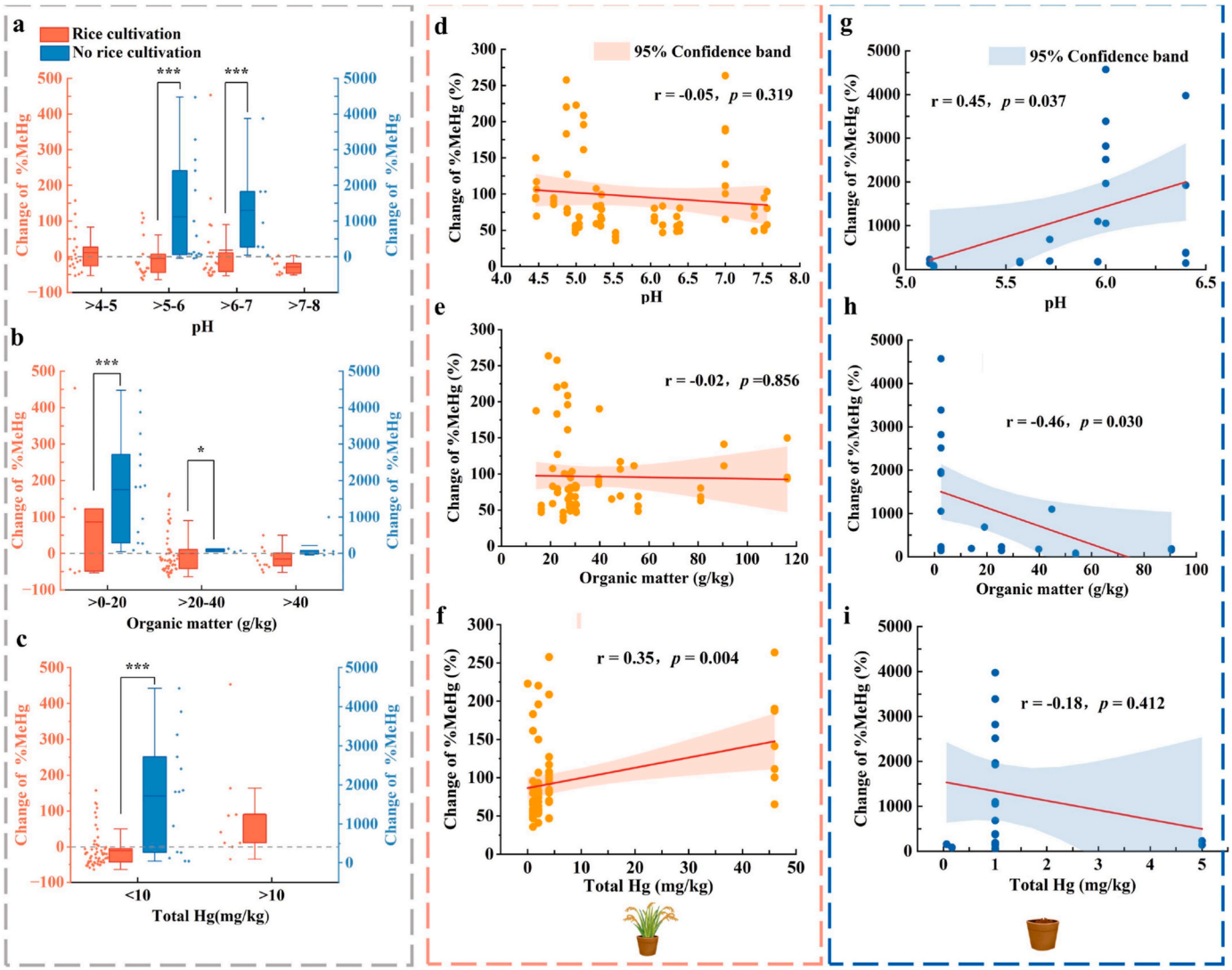


Fig. 3. Effects of pH (a), organic matter (b) and total mercury content (c) on mercury methylation in paddy soils under different cultivation conditions. The linear regression of change of soil %MeHg with pH (d), organic matter (e) and total mercury (f) under rice cultivation; The linear regression of change of soil %MeHg with pH (g), organic matter (h) and total mercury (i) under non-rice cultivation.

be related to the stability of MeHg in the soil. Increased pH values diminish the availability of both Hg and MeHg in the soil, as well as impair the soil's self-purification capacity, thereby facilitating the accumulation of these toxic elements in paddy soils (Gamboa et al., 2021; O'Connor et al., 2018; Yan et al., 2019). Furthermore, in the typical alternating dry and wet environment of paddy fields, Cu in the soil has the opportunity to increase the production of hydroxyl radicals (•OH) and superoxide radicals (• O_2^-) in an acidic environment, and these two types of radicals have been reported to directly promote the degradation of MeHg (Xie et al., 2019, 2020).

Intriguingly, MeHg accumulation exhibited a biphasic response to OM gradients, with suppression observed at both <10 g/kg and >60 g/kg OM contents in both paddy systems. This might be because when the OM is too low, Hg-methylation bacteria lack the electron donors needed to maintain metabolism, while when the OM is too high, they will complex with Hg (Yin et al., 2018). Sulfur-containing organic matter, particularly thiols, are strongly complexing ligands for Hg, as Hg binding to OM predominantly occurs through ionic reactions with naturally occurring sulfhydryl (•SH) functional group (Zhang et al., 2004). The binding affinity of Hg to sulfhydryl group is generally observed in the order: $(CH_3)_2Hg > Hg > CH_3HgX$ (where $X = OH^-$ or Cl^-) (Loux, 1998). A significant negative correlation between TS and %MeHg in the paddy fields of the Hg mining area further supported this mechanism ($p < 0.05$, Fig. S3a).

A recent study indicated that the diagenetic and decomposition processes of OM can inhibit Hg methylation and promote MeHg demethylation (Liu et al., 2022). It is worth noting that the optimal OM range for the accumulation of MeHg in Hg mining area paddy fields is slightly higher than that in ordinary paddy fields, which may be related to the composition and availability of OM (Bravo et al., 2017; Zhang et al., 2022). Our results also showed a weak negative correlation between TN and soil %MeHg (Fig. S3b). This is because that nitrogen can mitigate the promoting effect of carbon on MeHg (Li et al., 2023b). Meanwhile, elevated Hg contamination can enhance OM mineralization, thereby decreasing OM availability and reshaping microbial community composition (Macdonald et al., 2011; Skyllberg et al., 2006; Zhang et al., 2024). OM and %MeHg were significantly correlated only Hg mining area paddy fields (Fig. S4c), which also indicates that the differences in organic matter composition caused by paddy field types can affect the accumulation of MeHg. Therefore, we propose that remediation of MeHg-contaminated paddy fields may be achieved by lowering soil pH and strategically regulating OM content to suppress methylation potential.

4.2. Rice planting can affect the response of Hg methylation to environmental changes

We found that the paddy field soil has a relatively high Hg methylation capacity, and rice can significantly absorb MeHg in the soil (Fig. S5). However, the rhizospheric regulation of Hg methylation constitutes a multifactorial process involving root exudate chemistry, redox oscillations, and plant-microbe crosstalk. Firstly, certain components in the secretions of rice roots can promote the metabolism of Hg-methylation bacteria and facilitate the production of MeHg (Zhao et al., 2018). Secondly, rice secretes acidic substances that lower soil pH, thereby affecting microbial activity and the bioavailability of Hg (Rothenberg, 2016b), and the plant demethylation ability of rice is also worth noting (Strickman et al., 2017). Furthermore, a large number of studies have confirmed that rice roots can absorb Hg and MeHg from the soil and transfer them to rice, and this process is highly influenced by rice genotypes and specific environmental conditions (Abdelhafiz et al., 2023; Liu et al., 2022; Wiederhold et al., 2010). Meanwhile, the complex rhizosphere of rice provides convenient conditions for the attachment of microorganisms (Hang et al., 2025). Therefore, the impact of rice cultivation on soil Hg methylation is not yet complete, and more research is needed to improve the field of Hg pollution in paddy fields.

The effects of pH, OM and THg on Hg methylation in paddy fields were significantly affected by rice cultivation (Fig. S6), which might be due to the fact that rice roots would disturb the responses of Hg methylation and demethylation to environmental changes (Strickman et al., 2022). Furthermore, Acid biochar has been used to treat some soils contaminated by heavy metals (Esfandbod et al., 2017; Qi et al., 2018). Our results also indicated that the addition of acidic biochar may help limit Hg methylation pollution in acidic paddy soil. On the one hand, the addition of acidic biochar will further acidify the soil, resulting in the inhibition of the activity of some Hg-methylation functional microorganisms (such as iron-reducing bacteria, methanogens and sulfate-reducing bacteria) that are suitable for growth in neutral or weakly alkaline environments (Peng et al., 2024). Meanwhile, in a more acidic environment, free radicals with demethylation ability (such as •OH and $\bullet O_2^-$) is more likely to be produced, thereby promoting the non-biological demethylation of MeHg (Xie et al., 2020). On the other hand, the added organic matter can reduce the utilization of Hg by complexing with Hg to form Hg-OM complexes (Yin et al., 2018). Moreover, the thiol groups produced during the decomposition of some sulfur-containing organic matter can not only form Hg-S-OM complexes with Hg, but also promote the photodegradation rate of MeHg (Li et al., 2022). Furthermore, in the case of rice cultivation, the input of Hg should be restricted as much as possible. The large accumulation of Hg and MeHg can inhibit soil Hg methylation (Guo et al., 2021), but rice cultivation effectively alleviates this process, which may be due to the fact that rice roots transfer most of the Hg and MeHg in the soil to other tissues (Li et al., 2013). Therefore, it may also be a realistic option to select rice varieties with low accumulation of edible plant parts and to use plant accumulation capacity to rehabilitate Hg-contaminated paddy fields.

5. Conclusion

This study investigated MeHg contamination in paddy field ecosystem and found that MeHg levels were significantly higher in Hg mining area paddy fields compared to that in ordinary paddy fields. Among the physiochemical parameters examined, pH, OM and THg were the key determinants influencing MeHg production and accumulation. Meta-analysis results indicated that weakly alkaline environment (pH 7–8) favored MeHg accumulation. Both low and excessively high OM contents were unfavorable, with peak soil %MeHg occurring at OM levels of 20–30 g/kg in Hg mining area and 30–40 g/kg in non-mining sites. Furthermore, rice cultivation significantly absorbs a large portion of soil-generated MeHg, thereby modulating its accumulation. In acidic paddy field soils (pH = 5.0–6.5), reduction in pH combined with increased OM content may suppress Hg methylation. These findings contribute to a clearer understanding of how soil pH and OM influence MeHg production and accumulation in paddy fields, offering valuable insights for the development of remediation strategies targeting Hg-contaminated agricultural soils. we note that the scope of this meta-analysis is constrained by the availability and quality of published literature and data. Additional research is needed to elucidate the roles and mechanisms of other key factors-particularly C, N, and S in regulating MeHg dynamics within paddy soils.

CRediT authorship contribution statement

Lihu Ding: Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Yan Yang:** Methodology, Data curation. **Xuya Peng:** Conceptualization. **Lei Li:** Supervision. **Shu Yang:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition. **Liyan Song:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envres.2025.122358.

Data availability

Data will be made available on request.

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